

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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1. TITLE :

Design Static Webpage using HTML Components.

2. AIM :

To develop a static webpage using HTML components

3. OBJECTIVE :

The objectives of this experiment are:

- To design a static webpage using only core HTML components.
- To organize content using semantic HTML5 elements like `<header>`, `<nav>`, `<main>`, `<article>`, `<aside>`, and `<footer>`.
- To implement a multi-column layout using HTML tables for structural organization.
- To incorporate interactive form including text inputs, email fields, dropdown menus, textareas, and submit/reset buttons.
- To ensure accessibility through proper labeling, fieldset grouping, and semantic markup.
- To demonstrate internal anchor links for smooth page navigation.

4. THEORY :

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages using a system of tags and elements.

Semantic HTML refers to elements that clearly describe their meaning and purpose, such as `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<aside>`, and `<footer>`. These elements provide context about the content they contain.

Benefits of Semantic HTML:

- Accessibility for screen readers and assistive technologies
- Better SEO and search engine understanding
- Improved code maintainability and readability
- Consistent browser rendering across platforms

HTML Tables for Layout:

Tables (`<table>`) with rows (`<tr>`) and cells (`<td>`) are used to create structural layouts. Common attributes include width, border, cellpadding, cellspacing, align, and bgcolor.

Interactive Form Elements:

Forms collect user input using `<form>`, `<input>`, `<label>`, `<select>`, `<textarea>`, `<fieldset>`, and `<legend>`. Key attributes include action, method, required, and placeholder.

Hyperlinks:

The <a> tag enables navigation using href attributes for external links or internal anchor links (#id).

Browser Default Rendering:

Browsers apply default styles to all HTML elements, ensuring basic readability without any CSS.

Accessibility:

Proper semantic HTML ensures accessibility through label association, fieldset grouping, heading hierarchy, and descriptive link text.

Flow of a Static HTML Page:

Browser → Requests HTML → Server Serves File → Browser Renders → User Interacts.

5.ALGORITHM / PROCESS :

- 1 : Start the HTML document with <!DOCTYPE html> declaration.
2. Add the <html> tag with lang="en" attribute.
- 3: Create the <head> section and set character encoding using <meta charset="UTF-8">.
- 4: Set the page title using <title> tag.
- 5: Open the <body> section to add visible content.
- 6: Create a <header> element and add a table with website title and navigation links (Home, Articles, About, Contact Us).
- 7: Add a horizontal rule <hr> for visual separation.
- 8: Create a <section> element with id="home" for the hero section containing heading, paragraph, and a link to articles.
- 9: Add another horizontal rule <hr>.
- 10: Create a table with width="100%" for main content layout with two columns.
- 11: In the left column (width="70%"), add a <main> element with id="articles".
- 12: Inside <main>, create two <article> elements with headings, publication dates, and descriptive content.
- 13: Inside the second article, create a <form> with action="#" and method="post".
- 14: Add a <fieldset> with <legend> to group form elements.
- 15: Add form controls: text input, email input, dropdown select, and textarea with appropriate labels.
- 16: Add submit and reset buttons inside the form.
- 17: In the right column (width="30%"), add an <aside> element with id="about" for sidebar content.
- 18: Add unordered and ordered lists inside the sidebar.
- 19: Close the main content table and add a horizontal rule <hr>.
- 20: Create a <footer> element with copyright information.
- 21: Close all HTML tags properly (</body> and </html>).
- 22: Save the file with .html extension.
- 23: Open the file in a web browser to view the webpage.

26: Test form elements by entering data and clicking submit/reset buttons.

6. PROGRAM :

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>Pure Semantic HTML Static Webpage</title>

</head>

<body>

  <!-- header with the navigation menu-->

  <header>

    <table width="100%" border="0" cellspacing="0" cellpadding="10">

      <tr>

        <td>

          <h1>StaticWeb</h1>

        </td>

        <td align="right">

          <nav>

            <a href="#home">Home</a> &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp;

            <a href="#articles">Articles</a> &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp;

            <a href="#about">About</a> &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp;

            <a href="#contact">Contact Us</a>

          </nav>

        </td>

      </tr>

    </table>

  </header>

  <hr>

  <!-- Introduction Section -->

  <section id="home">

    <center>

      <h2>Semantic Structures Without CSS</h2>

    </center>

  </section>

</body>

</html>
```

<p>This layout uses raw HTML components to build an organized document hierarchy that browsers can cleanly parse out-of-the-box.</p>

<p>Read Our Articles Below ↓</p>

</center>

</section>

<hr>

<!-- Main Page Layout Divided into Content and Sidebar -->

<table width="100%" border="0" cellspacing="0" cellpadding="20">

<tr valign="top">

<!-- Left Side: Main Content Column -->

<td width="70%">

<main id="articles">

<article>

<h2>1. Core Structural Tags</h2>

<p>Published: July 7, 2026</p>

<p>When you omit cascading style sheets entirely, the semantic markup handles all of the structural heavy lifting. Elements like <code><main></code>, <code><article></code>, and <code><section></code> tell screen readers and web scrapers exactly how information flows.</p>

<p>Using raw browser defaults ensures excellent loading performance, perfect accessibility compliance, and absolute structural clarity.</p>

</article>

<hr>

<!-- Second Article with Form Demonstration -->

<article>

<h2>2. Native Interaction Elements</h2>

<p>Published: July 5, 2026</p>

<p>Browsers come equipped with excellent interactive elements by default. We can capture text inputs, provide multiple-choice configurations, and submit structured information to backend parsers using pure markup.</p>

<!--user interaction details-->

<h3>Interactive Demonstration Forms</h3>

<form id="contact" action="#" method="post">

<fieldset>

<!-- Grouping Related Form Fields -->

<legend>User Inquiry Form</legend>

<p>

```
<label for="name">Full Name: </label>

<input type="text" id="name" name="username" required placeholder="John Doe">

</p>

<p>

<label for="email">Email Address: </label>

<input type="email" id="email" name="useremail" required>

</p>

<p>

<!-- Dropdown Selection Field -->

<label for="topic">Inquiry Topic: </label>

<select id="topic" name="usertopic">

  <option value="general">General Feedback</option>

  <option value="technical">Technical Architecture</option>

  <option value="academics">Academic Curriculum</option>

</select>

</p>

<p>

<label for="message">Message Details:</label><br>

<textarea id="message" name="usermessage" rows="5" cols="40" placeholder="Type
your notes here..."></textarea>

</p>

<!-- Form Action Buttons -->

<p>

  <input type="submit" value="Submit Form Data">

  <input type="reset" value="Clear Entries">

</p>

</fieldset>

</form>

</article>

</main>

</td>

<!-- Sidebar Information Section -->
```

```
<td width="30%" bgcolor="#f4f4f4">
  <aside id="about">
    <h3>Document References</h3>
    <p>Static pages map data out directly without real-time database queries or processor
overhead.</p>
    <h4>Core Benefits:</h4>
    <ul>
      <li>Instant browser painting</li>
      <li>Low memory footprints</li>
      <li>High data compatibility</li>
    </ul>
    <h4>Required Components:</h4>
    <ol>
      <li>Document Type Definition</li>
      <li>Semantic Wrapper Blocks</li>
      <li>Data Input Control Interfaces</li>
    </ol>
  </aside>
</td>
</tr>
</table>
<hr>
<!-- adding the footer-->
<footer>
  <center>
    <p>&copy; 2026 Pure HTML Architecture. Assembled completely without style configurations.</p>
  </center>
</footer>
</body>
</html>
```

7. CODE EXPLANATION :

Code	Explanation
<!DOCTYPE html>	Declares the document type and HTML version.
<html lang="en">	Defines the root element with language set to English.
<meta charset="UTF-8">	Specifies character encoding for text rendering.
<title>	Sets the title displayed on the browser tab.
<header>	Represents the introductory section with logo and navigation.
<table>	Creates a layout for organizing content in rows and columns.
<tr>	Defines a table row.
<td>	Defines a table data cell.
<nav>	Defines a navigation section with hyperlinks.
	Creates hyperlinks for internal navigation.
<hr>	Adds a horizontal line for visual separation.
<section>	Defines a thematic grouping of content.
<main>	Specifies the primary content area.
<article>	Represents independent, self-contained content.
<form>	Creates a form to collect user input.
<fieldset>	Groups related form elements.
<legend>	Provides a caption for the fieldset.
<label>	Associates text with form controls.
<input type="text">	Creates a single-line text input field.
<input type="email">	Creates an email input field.

<select>	Creates a dropdown menu.
<option>	Defines options within a select menu.
<textarea>	Creates a multi-line text input area.
<input type="submit">	Creates a button to submit form data.
<input type="reset">	Creates a button to clear form entries.
<aside>	Defines sidebar content.
	Creates an unordered list.
	Creates an ordered list.
	Defines list items.
<footer>	Represents the footer section.
id	Assigns a unique identifier for anchor linking.
width	Sets the width of table columns.
align	Aligns content within table cells.
action="#"	Specifies the form submission URL.
method="post"	Defines the HTTP method for form submission.
required	Makes form fields mandatory.
placeholder	Displays hint text inside form fields.

8.TESTCASES AND EXECUTION DATA:

Test Case No.	Test Case Description	Input Data	Expected Output	Actual Output	Status
1	Verify webpage loads successfully	Open HTML file in browser	Webpage displays with all sections	Webpage loads with all sections	PASS
2	Verify navigation links	Click on "Articles" link	Scrolls to articles section	Scrolls to articles section	PASS
3	Verify form submission with valid data	Name: John, Email: john@email.com, Message: Hello	Form submits successfully	Form submits successfully	PASS
4	Verify form validation with empty fields	Name: (empty), Email: (empty)	Form prevents submission	Form shows validation error	PASS
5	Verify email validation	Email: "invalid"	Form shows validation error	Form shows validation error	PASS
6	Verify reset button functionality	Enter data and click reset	All fields clear	All fields clear	PASS
7	Verify articles section	View articles section	Two articles display with content	Two articles display with content	PASS
8	Verify sidebar section	View sidebar section	Lists and references display	Lists and references display	PASS
9	Verify footer section	View footer section	Copyright text displays	Copyright text displays	PASS
10	Verify semantic structure	Inspect HTML source	All semantic elements present	All semantic elements present	PASS

9. OUTPUT:

a) Header with navigation menu:

StaticWeb

[Home](#) | [Articles](#) | [About](#) | [Contact Us](#)

Semantic Structures Without CSS

This layout uses raw HTML components to build an organized document hierarchy that browsers can cleanly parse out-of-the-box.

[Read Our Articles Below .](#)

b)page layout : divided into content and sidebar

1. Core Structural Tags

Published: July 7, 2026

When you omit cascading style sheets entirely, the semantic markup handles all of the structural heavy lifting. Elements like `<main>`, `<article>`, and `<section>` tell screen readers and web scrapers exactly how information flows.

Using raw browser defaults ensures excellent loading performance, perfect accessibility compliance, and absolute structural clarity.

2. Native Interaction Elements

Published: July 5, 2026

Browsers come equipped with excellent interactive elements by default. We can capture text inputs, provide multiple-choice configurations, and submit structured information to backend parsers using pure markup.

Document References

Static pages map data out directly without real-time database queries or processor overhead.

Core Benefits:

- Instant browser painting
- Low memory footprints
- High data compatibility

Required Components:

- Document Type Definition
- Semantic Wrapper Blocks
- Data Input Control Interfaces

c)user interaction form:

Interactive Demonstration Forms

User Inquiry Form

Full Name:

Email Address:

Inquiry Topic:

General Feedback

Message Details:

Type your notes here...

Submit Form Data

Clear Entries

127.0.0.1

127.0.0.1:5500/index.html#

This page isn't working right now

If the problem continues, contact the site owner.

HTTP ERROR 405

After clicking submit:

d)footer:

10.RESULT:

The static webpage was successfully designed using pure HTML components without CSS or JavaScript. All semantic elements rendered correctly with default browser styles. The webpage displayed proper structure with header, navigation, hero section, articles, interactive form, sidebar, and footer. Navigation links functioned correctly, scrolling to respective sections. Form validation worked properly for required fields and email format. The reset button cleared all form entries successfully. All test cases passed, confirming that a well-structured static webpage can be created using only semantic HTML components.